

is a no-brainer.

4. There isn't any transmit filter definition in LTE, as is the case for 3G technologies. Hence in-channel performance (error vector magnitude) and out-of-channel performance (adjacent channel leakage ratio, spectrum emission mask) requirements need to be met by the design, which necessitates the testing process.

5. Testing multiple use of antennae (MIMO) as part of performance requirements is a must for a system which essentially uses that feature as a differentiator. Transmit diversity and spatial multiplexing (closed-looped and open-loop) are among the most important parameters to be tested.

Madhukar Tripathi, regional manager, Anritsu India, says, "The major parameters to be measured in LTE domain depend on the device under test (DUT). Peak data rate, control plane latency, user throughput and spectrum efficiency are measured for radio interface. Since LTE has different modulation schemes, those are also tested. The supported downlink data-modulation schemes are QPSK, 16QAM and 64QAM. The possible uplink data modulation schemes are (pi/2-shift) BPSK, QPSK, 8PSK and 16QAM."

Stephen Hire, general manager, Aeroflex Asia-India, adds, "While each generation of mobile phone technology brings greater benefits to the subscriber, it also adds greater complexity—especially in RF design, protocols and signalling, and battery life. This drives the need for ever more complex test equipment to bring mobile handsets and network equipment to market. Handset and chipset vendors are using our 7100 LTE radio test set and network emulator to optimise battery life and maximise data transfer rates to improve the user experience when browsing the Web or watching video. The larger the

LTE Test Solutions from Different Vendors

Vendors	Test solutions for LTE
Spirent	8100 LTE 8100 location technology
Aeroflex	TM500 LTE test mobile 7100 LTE UE tester
Rohde and Schwarz	FSW signal and spectrum analyser UPV audio analyser
National Instruments	LTE Measurement Suite
Agilent Technologies	N7109A multi-channel signal analyser W1715EP SystemVue MIMO channel builder
JDSU	DNA (J6801B) protocol analysis probe CellAdvisor JD745A and JD785A base station analysers
Interlab	LTE-USIM/USAT test solution
Ixia	IxCatapult Ixia BreakingPoint

network and the greater the number of subscribers, the more important it becomes to squeeze every last drop of performance out of each part of the network."

Tackling challenges

Being a relatively new technology, LTE poses unique challenges to test engineers. RamaRao Anil, product support and application, Rohde and Schwarz, explains, "LTE is fast changing, as we move between the different 3GPP releases. There is demand for new features to be incorporated in our test solutions. Also, these feature sets are quite new from the implementation perspective both for the chipset/UE and test vendors. Hence it becomes very important to collaborate and align roadmaps with the UE/chipset vendors to be successful."

Most LTE devices are expected to support both legacy 2G and 3G technologies. The testing has to take that into account. A host of valid scenarios for inter-technology mobility need to be validated to ensure seamless performance.

LTE itself offers two variants, viz, time-division LTE and frequency-division duplexing LTE, and separate testing is required for device flavours catering to each variant. Additionally, device vendors need to customise and maintain a large number of firmware

releases to address the varying needs of individual operators.

Sarma Yanamandra, wireless systems application engineer, Agilent Technologies, adds, "With the blistering speeds LTE offers, the major concern for the T&M vendors is with the HW to support it. MIMO is the way going forward for increased diversity and throughputs and there are several configurations that are defined. So, it becomes even more difficult to use the same HW to support such configuration due to

some obvious reasons. In addition, as LTE has to co-exist with the older technologies like 2G and 3G along with 1x and EvDo the T&M vendors have to feature the support in their products."

An LTE network is composed of multiple types of eNodeB's—femto, pico, metro and macro—each with a specific form factor and capacity. There is a clear need to focus not only on individual node types but also on how they stack up and work collectively, which is extremely essential as the collapse of the smallest layer can mess up the whole network. Also, functionality and performance of the 'brains' behind the success of heterogeneous networks (or HetNets), viz, self-organising networks and radio resource management, need to be carefully evaluated.

Ravichandran Raghavan, technical marketing engineer, National Instruments, India, summarises, "Traditional RF test and measurement vendors rely on existing tool-sets and methodology and try to extend it for newer communication standards, thereby not fulfilling the need for higher throughput and performance. With the ever increasing demand for testing LTE products, the number of LTE chip manufacturers is on the rise and there is a need for bringing down the cost and time to test these products and maintain a high quality."